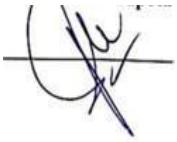
CardioAccess AI: An Multimodal Domain-Adaptive AI System for ECG and Echocardiography-Based Cardiac Screening

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Table of Contents

Abstract **4**

Introduction **5**

1.1 Background and Motivation 5

1.2 Problem Statement 5

1.3 Research Gap 6

1.4 Proposed Solution 7

1.5 Objectives 8

1.6 Scope and Limitations 10

1.7 Significance of the Study 10

Literature Review **11**

2.1 Introduction to the Review 11

2.2 ECG-Based Diagnosis and Arrhythmia Detection 11

2.3 Echocardiography and Domain Adaptation 11

2.4 Emerging Alternative Paradigms: Federated Learning and Foundation Models 12

2.5 Label Efficiency and Active Learning 12

2.6 Explainability and Fairness 12

2.7 Summary of Identified Gaps 13

2.8 Comparison Table 13

Proposed Work and its Methodology **16**

3.1 Dataset Acquisition (Seven total datasets). 17

3.2 Preprocessing. 17

3.3 Baseline Model Training. 18

3.4 Domain Adaptation and Active Learning. 19

3.5 Multimodal Fusion. 19

3.6 Model Compression 19

3.7 Explainable Prototype and Evaluation 19

3.8 Web Application Development and Deployment 20

System Architecture **22**

Software Development Methodology **23**

5.1 Planning 23

5.2 Development 23

5.3 Review 23

5.4 Retrospective 23
Applications of the Project 24
6.1 Frontline triage in specialist-scarce clinics: 24
6.2 Removing the specialist bottleneck: 24
6.3 Web-based access 24
6.4 Undergraduate / graduate medical & nursing training: 24
6.5 A reusable benchmark for future low-resource cardiac AI research: 24
Project Management 25
7.1 Workflow Chart 25
7.2 Project Breakdown 25
7.3 Gantt Chart 26
7.4 Roles & Responsibilities 27
7.4.1 Sahil Kumar: Data & ECG Modeling 27
7.4.2 Jawad Ali: Echo Modeling & Compression 27
7.4.3 Fatima: Integration & Product 27
7.4.4 Shared Responsibilities: 27
7.5 Tools & Technologies Required 27
7.6 Project Feasibility 29
Costing 30
Evaluation Metrics 31
Future Work 32
Conclusion 32
References 33

List of Figures

Figure 1: Literature Review Flowchart 9
Figure 2: System Abstract Model 16
Figure 3: Two-track Development Flowchart 17
Figure 4: System Architecture 22
Figure 5: Software Development Methodology Diagram 23
Figure 6: Workflow Chart 25
Figure 7: Gantt Chart 26
Figure 8: Cost Estimation 30

List of Tables

Table 1: Comparison Table 12
Table 2: Project Breakdown Table 20
Table 3: Tools & Technologies 22
Table 4: Costing 23
Table 5: Evaluation Metrics 24

Abstract

Cardiovascular disease has long been the leading cause of death globally, and it is now increasingly a problem for poor countries where they have limited access to heart doctors. Take Pakistan for example, which demonstrates the issue all too well the nation has barely 1,147 accredited cardiologists to serve a population of over 240 million, and there is no national approach to track or tackle heart disease. Artificial intelligence has been demonstrated to read heart signals and heart scans at levels of accuracy comparable to expert specialists, providing genuine hope in bridging this divide. Yet a panoramic view of the literature uncovers a pervasive and consequential limitation: the vast majority of these AI systems are developed and evaluated using data from a single institution or population, and the few that have undergone cross-institutional evaluation have been shown to experience a substantial degradation in performance when deployed elsewhere. A few recent initiatives have begun to address this generalization issue through (predictably) large-scale or computationally-expensive methods, but these solutions require infrastructure and data amounts that are not necessarily (or even often) feasible to get hold of. It is an open question whether a scaled-down, less expensive AI system would be as reliable across hospitals and patient populations and types of equipment.

This project seeks to fill that gap by direct examination. The goal is to design and experimentally validate a small form-factor dual-mode AI system for ECG analysis and echocardiography, and to perform extensive testing on how well such a system generalizes on patients and devices not used for training. Instead of accepting degradation in accuracy as the price to pay, this project will investigate feasible, low-cost options for reclaiming lost performance, with a special focus on how much realistic assistance in the form of new labeled data a resource-limited clinic would be able to receive in making the system better for use in its environment. It will also be programmed to attest when it is unsure about a prediction, so it can recommend that these cases be referred to a human rather than deliver a confident answer that might be wrong an important safety consideration for any tool designed to assist non-specialist healthcare professionals. Both will be combined into a single unified evaluation framework, and the developed system will then be rigorously tested not only in terms of accuracy but also efficiency on relatively low-cost hardware resembling what would be available in a primary-care facility. The end-product will be a working and explainable prototype, which will be piloted with primary care providers to determine if it is truly usable and trustworthy in real world.

Overall, this research aims to move beyond simply confirming that cardiac AI struggles to generalize, and instead to measure that problem precisely and demonstrate practical, affordable ways to reduce it, with the broader goal of making reliable AI-assisted cardiac screening genuinely achievable wherever specialist access is limited and clinical resources are constrained.

The deployed web application will also provide an on-demand Referral Summary Generation feature. After reviewing the completed screening results, an authenticated clinician can initiate the generation of a structured summary containing the available patient or case information, ECG and echocardiography findings, confidence and uncertainty estimates, and final triage outcome. The summary will be generated using a predefined template, stored securely with the corresponding case, and presented for clinician review before referral use.

Introduction

1.1 Background and Motivation

Cardiovascular disease (CVD) is the number one killer of people around the world and is not equally distributed; it is most prevalent in low and middle-income regions of the world where the medical system for diagnosis and treatment of heart disease is least developed. [1] Pakistan & Worldwide makes this crisis very real and concrete. The country in its entire history has only produced about 1,147 certified cardiologists to serve a population now over 240 million and there is no country-wide strategy in place to coordinate detection or risk management on a large scale.[2] The reality is that for a significant segment of the population, especially those in rural and semi-urban settings, there is little or no real opportunity for specialist cardiac assessment despite having apparent early signs and symptoms.

AI, especially deep learning, has become a technology that over the last decade has proven to be able to bridge just this type of gap. Deep learning models have repeatedly shown to perform at levels of expertise in two of the most common and clinically important cardiac diagnostic tasks: interpreting an electrocardiogram (ECG) for the detection of arrhythmias or ischemic events, and interpreting an echocardiogram for the assessment of cardiac structure and function, specifically the left ventricular ejection fraction (LVEF). Such models could enable a nurse or general physician at a resource-limited clinic to conduct a first-line cardiac screening which would normally be performed by a cardiologist. However, in practice, it is an open and important question whether this potential can be realized safely and reliably in a setting outside the particular hospital and population where a model was developed. [3] [4] [5]

1.2 Problem Statement

A systematic review was undertaken to determine if this concern is reasonable, rather than assumed, and to determine which of the major sub-areas of AI in cardiology (ECG diagnosis, echocardiography, heart failure risk prediction, wearable diagnosis and monitoring, and AI access in low-resource settings) there was a reasonable basis for concern. A structured review was conducted across over 60 studies published between 2019 and 2026, drawn from PubMed, Google Scholar, and IEEE Xplore, to determine if there was a reasonable basis for concern about the major sub-areas of AI in cardiology (ECG diagnosis, echocardiography, heart failure risk prediction, wearable diagnosis and monitoring, and AI access in low-resource settings). The trend revealed by this review was consistent and significant: Most published cardiac AI models were built and tested in a single institution, based on data from a single hospital, patients from a single population, and imaging or recording equipment from a single type. True cross-institutional validation of a model tested on data from a different hospital, country, or patient population than that on which it was trained is uncommon; when it is tried, its performance always and often drops significantly. This is not a hypothetical concern. Two recent, methodologically sound and solid studies make that point directly, not on the basis of assumption, but on the basis of real numbers. Gupta et al. (2026) developed the multimodal deep learning model (Conv-BiLSTM-Attn) for AMI detection using 145,656 ECGs from 96,813 patients from three hospitals in Sweden, and reported leave-one-hospital-out AUROC values from 0.801 to 0.849 showing that generalization is possible, but only once a large (expensively-labeled) multi-institutional dataset has been assembled. [6] For cross-domain

echocardiographic segmentation, Nazari et al. (2025) introduced an unsupervised framework that combines a Variational AutoEncoder, Wasserstein GAN and a dual-attention network, achieving a Dice score of 84.6% and 89.1% on the CAMUS and EchoNet-Dynamic databases respectively, and thus pointing out the possibility of generalization with a rather heavy adversarial training architecture. [7] None of the results show that this robustness can be obtained under the compute and data and labeling constraints that are actually faced in a primary care clinic in a country like Pakistan, but both results show that the underlying generalization problem is real and solvable in principle.

1.3 Research Gap

Taken together, the literature review point to seven specific, unaddressed gaps, each of direct practical consequence to a resource-limited clinic:

1. Cross-modal robustness to domain shift is untested**:** Both Gupta et al. and Nazari et al. consider a single modality independently (ECG and echocardiography, respectively). None investigate if a combined use of both modalities can yield a Triage Decision more stable in its prediction to domain shift than any single modality taken alone. [6] [7]

    - This gap extends further to the equipment level: neither study evaluates performance under single-lead ECG acquisition, added signal noise, or reduced echocardiographic video quality the actual conditions of low-cost portable devices in primary-care settings, as distinct from cross-institutional differences alone

2. Computational complexity and hardware implementability are not reported: The latency, the memory usage, and the cost of inferring of the proposed methods on low-power hardware representative of a primary-care setting are not evaluated in either work, considering that the architecture of Nazari et al. contains efficiency oriented design elements such as depth wise separable convolutions. [7]

    - This limitation extends to the adaptation process itself: it remains unknown whether the adaptation methods used to achieve cross-institutional robustness including Nazari et al.'s adversarial training pipeline can feasibly run on constrained hardware, or whether adaptation structurally requires centralized compute regardless of how efficient the final deployed model is.

3. Confidence calibration under domain shift is not studied: Neither study probes into whether a model's own confidence predictions would be really dependable when used outside of its own training distribution a model can be inaccurate and confidently wrong without the evaluation protocol in either paper detecting that.

    - A related and equally unexamined question is whether the process of adaptation causes a model to increasingly rely on demographic-correlated signal patterns rather than genuine pathological features, meaning a model could become well-calibrated while quietly shifting what it is actually keying on.

4. Label-efficient adaptation is not tested**:** The robustness of Gupta et al. comes from the scale of their training data, not from adaptation; Nazari et al.'s unsupervised domain adaptation framework, while requiring no labeled target data, still depends on access to the *entire* available target-domain dataset during training rather than being evaluated under a deliberately limited, clinic-realistic sample size. Neither study, therefore, establishes how much labeled or unlabeled target data is actually required to achieve robust cross-institutional

performance, or whether adaptation quality degrades gracefully or collapses sharply as available local data shrinks toward what a genuinely resource-limited clinic could realistically provide. [6] [7]

5. Dominated by single-institution validation: Almost all (≥90%) of published cardiac-specific models (ascertained among 60+ citable studies, 2019-2026) are trained and tested using data from a single institution or dataset, with true external validation not being the standard but the exception.
6. Active learning has not been applied to cardiac signals or images**:** Active domain adaptation using a model's own uncertainty to select which unlabeled samples are most valuable to label has been demonstrated in other medical imaging contexts, including tumor segmentation and general federated image analysis, but has not been tested on ECG or echocardiography, where signal noise and variability characteristics differ substantially from the imaging tasks where this technique has previously been validated.
7. Neither comparator study includes any human-factors or usability evaluation**:** Both Gupta et al. and Nazari et al. are purely technical benchmarking studies; neither assesses whether a non-specialist clinician would understand, trust, or correctly act on the model's output, leaving the practical usability of these approaches in a real clinical workflow entirely untested.

1.4 Proposed Solution

This project proposes CardioAccess AI, a domain-adaptive, multimodal, uncertainty-aware deep learning system for cardiac screening, designed explicitly to investigate and address these seven gaps rather than to compete on raw accuracy with either comparator study. Two compact models are developed: an ECG arrhythmia classifier trained on PTB-XL [19] and externally validated on ARRDB [11] and a locally sourced dataset, and a segmentation-based EF estimator with a reduced-EF screening output trained on EchoNet-Dynamic [3] and externally validated on CAMUS [8]. Both models are subjected to domain adaptation (limited-data fine-tuning, correlation alignment, and test-time adaptation) [9], with target-domain sample selection guided by an uncertainty-guided active learning strategy built on Monte Carlo dropout and deep ensemble confidence estimates [10]. The same confidence estimates are further evaluated for calibration before and after domain shift and used to flag low-confidence predictions to the end user rather than presenting false certainty. The two modalities are combined through a learned multimodal fusion network, benchmarked against both a rule-based baseline and each single-modality adapted model, to test directly whether joint reasoning improves cross-domain robustness. Both final models are compressed via quantization and knowledge distillation and benchmarked for accuracy, latency, and memory trade-offs on low-end hardware representative of a primary-care setting. These components are integrated into a working, explainable web application, evaluated through a structured usability pilot with primary care providers. MIMIC-IV-ECG [21] and MIMIC-IV-Echo [22] will additionally provide candidate ECG and echocardiography records for multimodal fusion, using only records with verified patient-level linkage and suitable temporal correspondence between the two modalities.

Beyond the functionalities of screening and result visualization, proposed web application would also include an On-Demand Referral Summary Generation feature. Allowed by viewing the completed analysis in the dashboard, the function can be triggered by an authenticated clinician. The FastAPI backend will download available case information and archived ECG, echocardiography and multimodal analysis outputs where relevant, and will arrange everything into prestructured summary. The summary produced will then be saved in Supabase associated with the respective case and will be available for review by the clinician. Instead of creating a second diagnosis or altering model predictions, this feature will compile existing screening documentation.

1.5 Objectives

The specific objectives of this project are to:

- Conduct a structured literature review of AI applications in cardiology from 2019–2026, with particular emphasis on ECG-based arrhythmia classification and echocardiography-based ejection-fraction estimation, to identify the extent and consistency of cross-institutional external validation.

- Develop and train an ECG-based arrhythmia classification model and quantify its cross-institutional generalization gap through external validation on a dataset from a different source or institution.

  This combines:
  - ECG development
  - ECG external validation
  - generalization-gap measurement
  - segmentation, EF estimation, reduced-EF screening, external validation, generalization-gap measurement.

- Develop and train a video-based echocardiographic segmentation and ejection-fraction estimation model and quantify its cross-institutional generalization gap through external validation on an independently sourced dataset.

  This combines:

  - Echo segmentation
  - EF estimation
  - external validation
  - generalization-gap measurement

- Evaluate lightweight domain adaptation strategies, including limited-data fine-tuning, correlation alignment, and test-time adaptation, to determine their effectiveness in recovering cross-institutional performance loss in both ECG and echocardiography models while quantifying the associated labeled-data requirements.
- Implement an uncertainty-guided active learning strategy to select informative local samples for adaptation and compare its labeling efficiency against random sample selection.
- Evaluate model confidence calibration under cross-institutional domain shift and determine whether calibration is preserved or degraded before and after domain adaptation. For the echocardiography model, calibration is computed on the reduced-EF probability, since calibration requires a classification probability. This is a separate research question.

  You can measure things such as:
  - Expected Calibration Error (ECE)
  - Brier score
  - reliability diagrams

- Design and evaluate a learned multimodal fusion model combining ECG and echocardiography predictions, and compare its robustness to domain shift against rule-

based fusion and single-modality models.

- Compress the final ECG and echocardiography models using quantization and knowledge distillation, and benchmark their accuracy, latency, memory usage, and computational feasibility on hardware representative of a resource-limited primary-care setting.
- Integrate the validated models, adaptation, uncertainty, calibration, fusion, and explainability components into a working prototype, and evaluate its usability and perceived trustworthiness through a structured pilot involving non-specialist healthcare workers or primary-care providers.
- Implement an on-demand Referral Summary Generation feature that enables authenticated clinicians to generate, review, and securely store structured summaries containing available patient or case information and completed cardiac screening results, and evaluate its completeness, consistency, and usability.

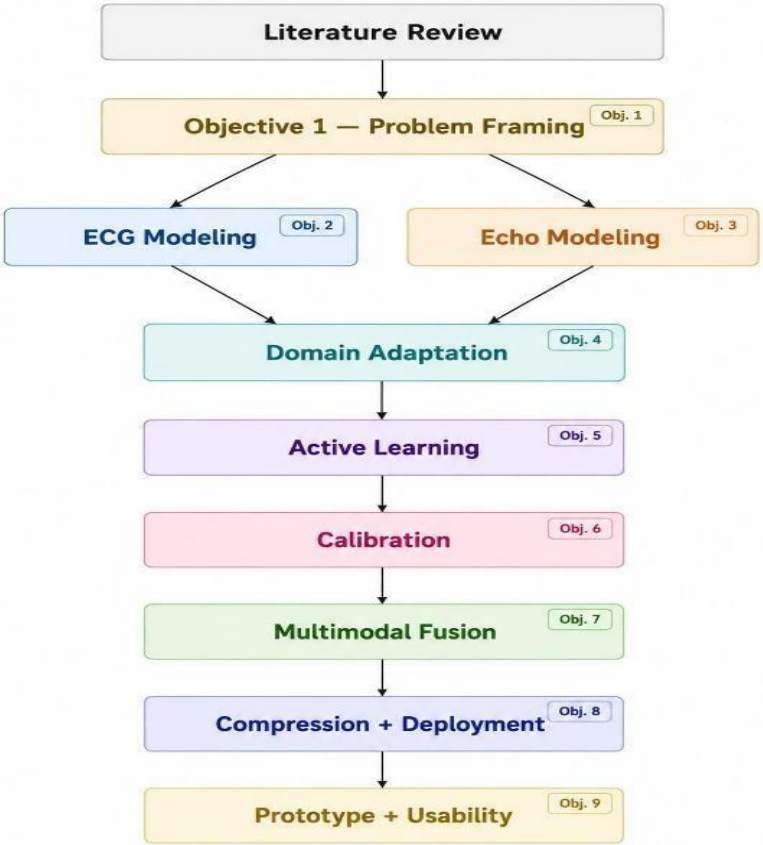


Figure 1: Literature Review Flowchart

1.6 Scope and Limitations

This project is scoped as a research prototype and feasibility study, and not a clinically validated or regulatory-cleared medical device. Model development will use seven de-identified research datasets: PTB-XL [19], ARRDB [11], CPEIC (Multan Dataset) represents the locally sourced ECG dataset and is counted as one dataset. [20] , EchoNet-Dynamic [3], CAMUS [8], MIMIC-IV-ECG [21], MIMIC-IV-Echo [22], and a locally sourced ECG dataset (seven datasets total). MIMIC-IV resources are access-controlled and require the applicable credentialing and data-use approval. No primary patient data is collected, and the usability evaluation is a simulation study with volunteer participants, explicitly not a clinical trial. The reproduction of Gupta et al.'s and Nazari et al.'s techniques (see Section 1.3) is a smoke test to ensure that implementation is possible, not a claim of reproducing their full trained reported results [6] [7].

The web app will also allow for on-demand structured referral-support summaries to be generated after the relevant screening analysis has been completed. Summaries will be created from a template using pseudonymous case information and existing model outputs. The prototype will use simulated case information that is de-identified or no patient data is collected for data mining use. Action to generate summary will necessitate clinician access token authentication, and the summary will be oranges for clinician review. The summaries are not diagnoses, treatment plan, or validated medical reports but a blend of screening findings among which we can consider for further screening.

1.7 Significance of the Study

By directly building upon, incorporating aspects of, and strengthening the two most recent and methodologically sound published approaches to cardiac AI generalization, this work elevates the focal question in this domain from is cross-institutional robustness achievable in principle, which both comparator studies already demonstrate to can it be achieved, cost-effectively, and in a manner that is repeatable and trustworthy given the computational, data, and labeling constraints that actually characterize a resource limited clinic. This refocusing has immediate implications for the healthcare system in Pakistan, where the specialist access gap outlined in Section 1.1 is both severe and well-known, as well as in similar resource-strapped regions globally.

The On Demand referral summaries also help enhance the practical value of the prototype by compiling whatever screening information is available and triage info into a format that can facilitate communication with specialist healthcare providers, in a structured manner.

Literature Review

During this investigation, a comprehensive and stringent literature review was being written based on 60 papers found available on the platforms of PM, GS, extraction tables constructed, gap statements developed, and two closely related papers were extensively analyzed but this work has been existing in research notes and in the Summary paragraphs of the Introduction chapter and not yet as its own separate, structured chapter. Below is that chapter, in its entirety.

2.1 Introduction to the Review

In this chapter, we described the state-of-the-art of artificial intelligence in cardiology in 2019–2026 focusing on two modalities that are at the core of this project: the diagnosis based on electrocardiogram (ECG) and the evaluation of cardiac function by echocardiography. The review is based on more than 60 papers selected from the databases PubMed, Google Scholar and IEEE Xplore and is thematically structured along the five research gaps that provide the motivation of this project.

2.2 ECG-Based Diagnosis and Arrhythmia Detection

ECG deep learning methods have evolved from classical feature-based methods (support vector machines run on hand-engineered waveform features) towards end-to-end convolutional and recurrent architectures that can achieve cardiologist level performance on arrhythmia classification tasks. Work such as Hannun et al. (2019) has long demonstrated large scale ECG classification on ambulatory recordings, [5] while benchmarking analysis on the PTB-XL dataset (Strodthoff et al., 2021) revealed that ResNet and Inception-based one-dimensional convolutional architectures come at the top of classical approaches across diagnostic subtasks. [4] An all-too-frequent aspect and well-known pitfall of this literature is the potential for overestimated performance due to incorrect train/test splits: the CPSC2018 challenge documentation explicitly cites one instance where accuracy dropped from 99.7% to 81.5% when evaluation was fixed from an intra-patient to an inter-patient split, clear evidence that many reported high-accuracy results are in fact memorizing patient-specific signal features rather than truly diagnostic-generalizing. [12] Most recently, and most directly relevant to this project Gupta et al. (2026) introduced the Conv-BiLSTM-Attention model for acute myocardial infarction detection, trained on 145,656 ECGs from three real Swedish hospitals, thoroughly tested both in pooled cross-validation and leave-one-hospital-out external validation demonstrating AUROC scores of 0.80–0. 85 that were fairly consistent among the different sites a rare, believable instance of large-scale, real hospital data resulting in truly generalizable performance, derived through data volume and variety rather than post-hoc adaptation. [6]

2.3 Echocardiography and Domain Adaptation

Automated echocardiographic analysis, particularly left ventricular segmentation for ejection fraction estimation, has been substantially advanced by large public datasets, notably EchoNet-

Dynamic (Ouyang et al., 2020), which demonstrated video-based deep learning achieving expert-comparable segmentation (Dice 0.92) and ejection fraction estimation (mean absolute error 4.1%), with confirmed external validation on a separate health system. [3] The CAMUS dataset (Leclerc et al., 2019) established a complementary benchmark using French hospital data, with U-Net-based architectures achieving Dice scores above 0.93 [8] .

The specific challenge of cross-institutional and cross-equipment domain shift in echocardiography has been directly addressed by GraphEcho (2023), which demonstrated that segmentation performance can collapse from 81.3% to 48.5% Dice when a model is applied across sites without adaptation, and proposed a graph-driven unsupervised domain adaptation solution [13]. Most recently, Nazari et al. (2025) advanced this further with a framework combining a Variational AutoEncoder, a Wasserstein GAN for domain alignment, and a dual-attention mechanism, achieving Dice scores of 84.6–89.1% in bidirectional adaptation between CAMUS and EchoNet-Dynamic, and extending validation to a third, previously unpublished local hospital dataset from Iran directly paralleling the role a local Pakistani dataset would play in the present project, though achieved through a computationally intensive adversarial architecture rather than lightweight adaptation. [7]

2.4 Emerging Alternative Paradigms: Federated Learning and Foundation Models

Beyond direct domain adaptation, two other paradigms have emerged to address cross-institutional generalization. Federated learning, in which models are trained collaboratively across institutions without centralizing raw patient data, has been applied successfully to multimodal ECG and echocardiography-based hypertrophic cardiomyopathy detection (Goto et al., 2022), achieving C-statistics of 0.90–0.96 across multiple countries [14], and validated at very large scale (1.5 million ECGs across seven Canadian hospitals) using federated learning combined with differential privacy [15]. Separately, self-supervised foundation models pretrained on large, diverse datasets have begun to be evaluated explicitly for cross-site generalization in echocardiography, exemplified by the 2026 CardioBench benchmark, which tests multiple foundation models across eight public datasets [16].

Both paradigms confirm that cross-institutional robustness is achievable, but both require infrastructure either multi-institutional data-sharing coordination (federated learning) or large-scale pretraining compute (foundation models) that is generally unavailable to an individual resource-limited clinic, motivating the present project's focus on lightweight, locally-adaptable alternatives.

2.5 Label Efficiency and Active Learning

A separate but related thread in the broader medical imaging literature addresses the cost of labeling data for adaptation. "Active domain adaptation" combining domain adaptation with active learning-based selection of the most informative samples to label has been demonstrated in other medical imaging contexts, including source-free active domain adaptation for tumor segmentation across multiple hospitals, [17] and federated evidential active learning frameworks. [18] No identified study applies this combined approach to cardiac ECG or echocardiography specifically, representing a direct, addressable gap for this project.

2.6 Explainability and Fairness

Model interpretability has become an increasingly expected component of clinical AI research. ScoreCAM and Grad-CAM-style visualization techniques have been used to confirm that ECG

models attend to physiologically meaningful regions such as the ST segment and T wave (Gupta et al., 2026), lending clinical plausibility to model predictions. Separately, documented evidence of demographic under diagnosis bias in chest imaging AI, and the general scarcity of subgroup-stratified performance reporting across FDA-authorized AI/ML medical devices, motivates the inclusion of explicit fairness auditing in this project's evaluation plan.

2.7 Summary of Identified Gaps

Synthesizing the above, seven gaps directly motivate this project: the continued dominance of single-institution validation across the field at large; the absence of any study testing whether cross-modal (ECG plus echocardiography) fusion improves robustness to domain shift; the absence of computational cost or hardware feasibility reporting in the two closest comparator studies; the absence of confidence calibration evaluation under domain shift; and the absence of label-efficient, actively-selected adaptation specifically applied to cardiac signals or images; the absence of any test of whether active-learning-guided sample selection functions on cardiac ECG or echo data specifically; and the absence of any human-factors or usability evaluation in either comparator study. These seven gaps, detailed further in Chapter 1's Research Gap section, define the scope of the work undertaken in this project.

2.8 Comparison Table

Table 1: Comparison Table

#	Study	Component	Model/Architecture	Dataset(s)	Key Reported Result	Limitation / Trade-off
1	Strodthoff et al. (2021)	ECG	XResNet1d101 (plain 1D CNN)	PTB-XL (in-distribution)	Macro AUC ≈ 0.93–0.975	Strong in-distribution accuracy; no cross-institution validation or hardware efficiency reporting
2	Hannun et al. (2019)	ECG	Deep 1D CNN (34-layer residual)	Single-institution ambulatory ECG (91,232 recordings)	Cardiologist-level arrhythmia classification	Large-scale but single-institution; no external validation
3	Gupta et al. (2026)	ECG	Conv-BiLSTM-Attention	3 Swedish hospitals (145,656 ECGs)	AUROC 0.80–0.85 (leave-one-hospital-out)	Genuine cross-hospital validation; recurrent layer adds inference cost, not edge-benchmarked
4	Liu et al. (2018) CPSC2018	ECG	Various CNN baselines	CPSC2018 (intra- vs. inter-patient split)	Accuracy fell from 99.7% to 81.5% on corrected split	Evidence of generalization overestimation, not a proposed architecture
5	Ouyang et al. (2020) EchoNet-Dynamic	Echo	DeepLabv3 + R2+1D 3D-CNN	EchoNet-Dynamic + external health system	Dice 0.92; EF MAE 4.1%; AUC 0.97	Two heavy models; no hardware efficiency benchmarking

6	Leclerc et al. (2019) CAMUS	Echo	U-Net (best of 4 benchmarked)	CAMUS (single-center, France)	Dice 0.939 (ED) / 0.916 (ES)	Single-institution only, no cross-institution test
7	Yang et al. (2023) GraphEcho	Echo	Graph-driven unsupervised domain adaptation	CAMUS ↔ EchoNet-Dynamic	Dice collapses 81.3%→48.5% without adaptation; recovered with UDA	Confirms the domain-shift problem; graph-based UDA is computationally heavy
8	Nazari et al. (2025)	Echo	VAE + Wasserstein GAN + dual-attention	CAMUS, EchoNet-Dynamic, Heshmat (Iran)	Dice 84.6–89.1% (cross-domain); 83.0–84.1% (local hospital)	Strong cross-country result; adversarial training is unstable and heavy, no hardware benchmarking
9	Cardiac amyloidosis detection study (multimodal fusion)	ECG + Echo (Fusion)	Transformer-based fusion (ECG selects high-risk patients, echo model gives final prediction)	Single-institution paired ECG-echo cohort	AUROC 0.96 (amyloidosis detection)	High accuracy on one specific rare disease; single institution, not tested under domain shift
10	Multimodal foundation model (2025)	ECG + Echo (Fusion)	Self-supervised contrastive joint embedding	Large paired ECG-echo cohort	Strong performance across multiple downstream cardiac tasks	Foundation-model scale (heavy pretraining), not benchmarked for edge deployment or label efficiency
11	**Proposed (this study)**	**ECG**	**SE-ResNet1D** (1D CNN + Squeeze-and-Excitation channel attention), or Fallback existing architecture	PTB-XL (train); ARRDB and local dataset (external validation)	Target: AUC ≥0.90 in-distribution, with cross-dataset generalization, parameter-count, and latency reporting	Results pending Semester 1 completion

12	**Proposed (this study)**	**Echo**	**Lightweight U-Net** (MobileNetV3/ EfficientNet-B0 encoder) + volume-curve EF regression head or existing architecture	EchoNet-Dynamic (train); CAMUS (external validation)	Target: Dice ≥0.85 in-distribution, with latency/memory benchmarking on constrained hardware	Results pending Semester 1 completion
13	**Proposed (this study)**	**Fusion**	Multimodal Learned Fusion (ECG & Echo)	MIMIC-IV-ECG and MIMIC-IV-Echo (paired records, subject to linkage verification)	Target: measurable AUC improvement over either single-modality model, with interpretable gate weights showing modality reliance under domain shift.	Results pending Semester 2 completion
14	**Proposed (This study)**	**Application**	Fully deployable React (Vite + Tailwind CSS) web application + Supabase PostgreSQL Database, Authentication and Row-Level Security (RLS) + FastAPI/ONNX Runtime inference service	De-identified 12-lead ECG and echocardiography datasets, including patient-paired MIMIC-IV ECG–Echo records for multimodal fusion	Target: ECG classification (AUC, sensitivity, specificity), echo segmentation and EF estimation performance, multimodal fusion performance, functional and security testing, and SUS-based usability evaluation.	Results pending final-semester completion

Proposed Work and its Methodology

Overall Conceptual Model of CardioAccess AI

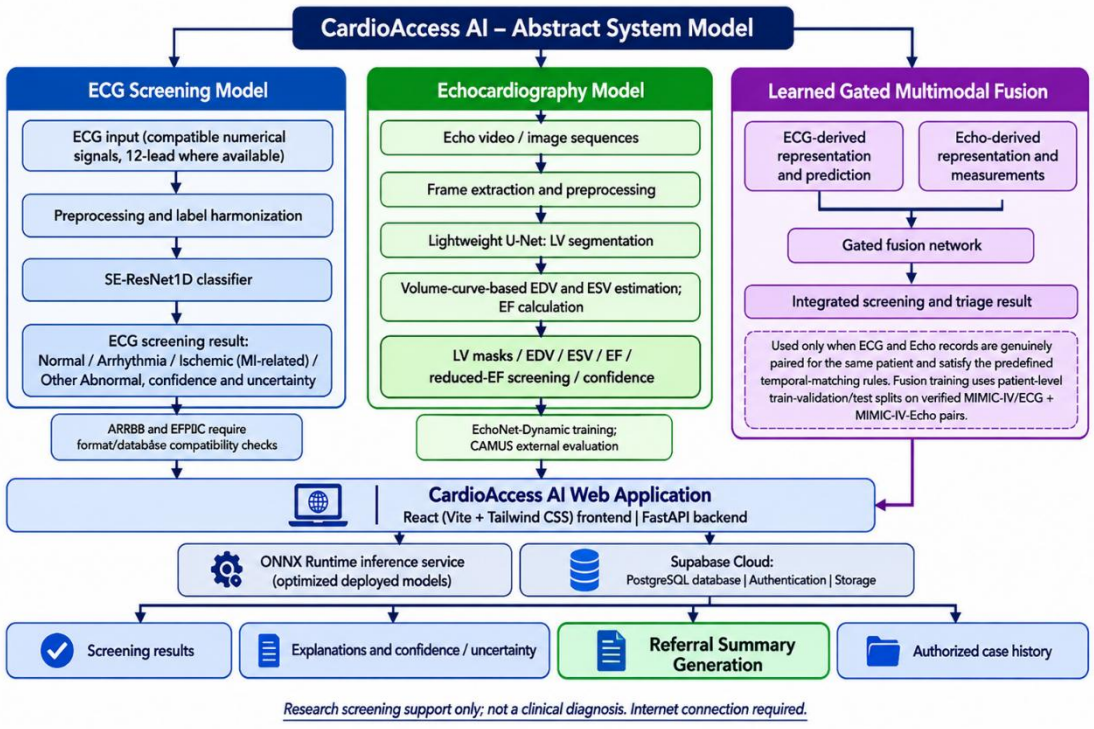


Figure 2: System Abstract Model

CardioAccess AI is a multimodal cardiac screening system that uses 12 lead ECG signals and echocardiography data to detect possible cardiovascular abnormalities. The first model employs ECG screening wherein it uses selective datasets to amalgamate the diagnostic categories by preprocessing, classification, and domain adaptation. The second uses a lightweight U-Net to analyze an echocardiography of the frames (video or a set of. pngs) and identifies the boundary of the left ventricle via segmentation. Given the ventricular regions found via segmentation, it measures End-Diastolic Volume (EDV), End Systolic Volume (ESV) and Ejection Fraction (EF) to monitor reduced function. For the third model, we use ECG and echocardiography records that are truly patient-paired with one another from MIMIC-IV and model the multimodal data via a learned gated fusion network. This fusion model is able to produce confidence aware screening and triage such as Normal, Needs Review, Urgent Referral while routing doubtful cases for a second opinion. Visual and confidence-based insights into the model predictions via explainable AI techniques, which was developed to assist the users to interpret results in whole system outputs. At the end, we could deploy all of the optimized models as a complete web application (powered with React, FastAPI, and ONNX Runtime) as we use Supabase for the entire authentication, secure database storage stack - all are online, scalable, and easily accessible for screening.

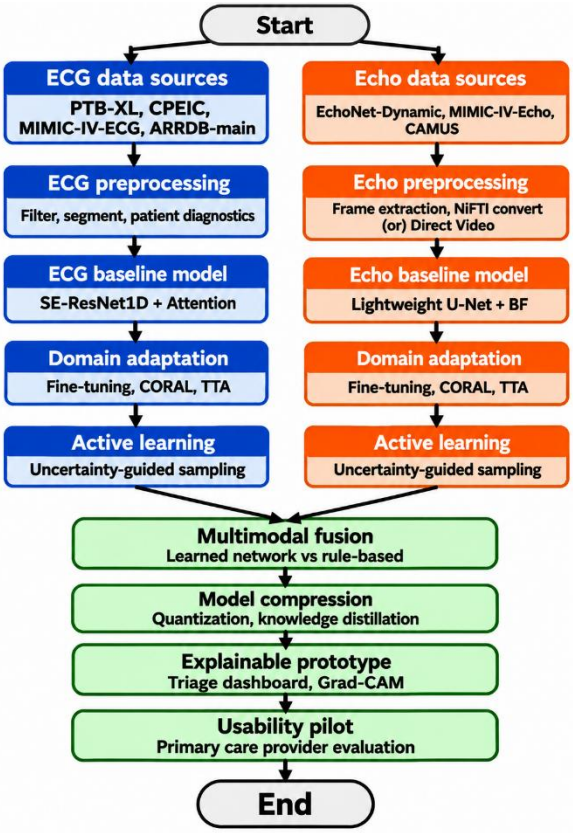


Figure 3: Two-track Development Flowchart

The two-track flowchart above illustrates the full development pipeline. In summary form, matching the reference's step-by-step explanation style:

3.1 Dataset Acquisition (Seven total datasets).

For the ECG track: MIMIC-IV-ECG [21], PTB-XL [19], the VitalDB Arrhythmia Database (ARRDB; 482 surgical patients and over 660,000 validated beat/rhythm annotations) [11], CPEIC (the Multan lineage dataset) [20]. The exact identity and access information of the local dataset should be documented precisely (source institution, size, label format, and ethical-clearance status) before it is accepted for training or validation. For the echo track: MIMIC-IV-Echo [22], EchoNet-Dynamic [3], and CAMUS [8]. Together, these constitute seven datasets across the two modality tracks. MIMIC-IV data access is credentialed and subject to the applicable data-use requirements.

3.2 Preprocessing.

ECG signals are filtered, segmented, and split at the patient level with no data leakage; local/private datasets are converted to a unified label taxonomy based on the diagnostic classes of PTB-XL. Data from Echo needs to be get directly video or extracted from each frame of the EchoNet-Dynamic video format and CAMUS's NIfTI sequences must be converted to a similar format of representation. CPEIC is image-based and must be converted into numerical ECG signals before being processed alongside signal-based ECG datasets.

- Sample rate normalization: Choose a single target rate (250 Hz) and resample compatible ECG datasets to this rate during preprocessing with a standard resampling function. This

is implemented in each dataset loader and documented in the Methodology chapter.

- 12-lead standardization: Retain all 12 ECG leads as model inputs, harmonize their channel ordering and signal representation across compatible datasets, and resample the signals to a common target frequency of 250 Hz. Any missing or incompatible leads will be documented and handled using a consistent strategy.

- Label taxonomy reconciliation: Construct a single table, feature it as a deliberate methodological artifact. Establish a relatively small shared output space e.g. in 4 shared categories: Normal / Arrhythmia / Ischemic (MI-related) / Other Abnormal and create an explicit mapping table that indicates exactly which of the original label in each data set is mapped to which of the shared category. This table becomes an actual, citable artifact in your Methodology chapter (a real contribution reviewers respect, since most papers hide this decision rather than document it). If a dataset's granularity doesn't map cleanly, report your primary cross- dataset comparison at the coarsest shared granularity, and anything finer grained just within that one dataset.

- There are existing published, customizable methods for ECG image to signal digitization (grid detection, trace extraction, pixel-to-voltage calibration based on known paper speed/gain). Just adopt one instead of reinvent the wheel. Once digitized, if digitization is too time-consuming (which, considering your Semester 1 timetable, it very well might be), the well-honest backout (also zero architecture change) is to have CPEIC be a named additional experiment rather than mashing it quietly into your main comparison this is a scope decision that you disclose, not a breach of your principle design.

- Generate binary segmentation masks by connecting the 20 left-side and 20 right-side boundary points of each annotated frame to form a closed polygon. Label the larger ventricular area as End-Diastole (ED) and the smaller area as End-Systole (ES).

3.3 Baseline Model Training.

The ECG model is an SE-ResNet1D architecture a one-dimensional convolutional residual network enhanced with Squeeze-and-Excitation channel-attention blocks selected in preference to the recurrent Conv-BiLSTM-Attention architecture used by Gupta et al. (2026). [6] since a fully convolutional design avoids the sequential inference cost of recurrent layers and is better suited to quantization and edge deployment. As a direct comparison, a standard ResNet1D and Inception1D (the architecture benchmarked in Strodthoff et al. [4] are also trained as baseline ablations, to quantify the accuracy-versus-efficiency trade-off explicitly. The echo model is a Lightweight U-Net, using a MobileNetV3 or EfficientNet-B0 encoder in place of a full-scale segmentation backbone, in preference to the heavier DeepLabv3 and R2+1D 3D-CNN architecture used by EchoNet-Dynamic. [3] Ejection fraction is estimated via a shallow volume-curve regression head applied to the segmentation output across the cardiac cycle, rather than a direct video-to-value regression network, favoring an interpretable, clinically-motivated pathway. The echo pipeline produces three connected outputs. A compact U-Net first segments the left ventricle (LV) in every video frame. The area of these masks is then used to create a volume curve over the cardiac cycle, from which end-diastolic volume (EDV, the maximum) and end-systolic volume (ESV, the minimum) are estimated. Ejection fraction is calculated as EF = (EDV − ESV) / EDV × 100. The estimated EF also supports screening for reduced function. The main cutoff is EF < 50%, while EF ≤ 40% is used as a secondary threshold to flag severe dysfunction; both reflect standard clinical categories. For each cutoff, the probability of reduced EF is estimated from the EF prediction distribution generated by Monte Carlo dropout and an ensemble. Training masks come from expert LV tracings in EchoNet-Dynamic. Because annotations are available only for the end-diastolic and end-systolic frames, the model is trained on those labeled frames and then applied to every frame during inference. A direct video-to-EF regression model is also trained as a baseline ablation.

3.4 Domain Adaptation and Active Learning.

Both models are fine-tuned on external datasets with limited data, alignment of correlations (CORAL), and adaptation time testing [9], with selection of sample targets being guided by scores of uncertainty (Monte Carlo dropout, deep ensembles) [10] as opposed to being random. For the echo model, adaptation is evaluated on all three outputs: segmentation quality (Dice), EF estimation error (MAE), and reduced-EF screening performance (AUC), so that recovery after adaptation is reported separately for each output rather than as a single aggregate number [17][18].

3.5 Multimodal Fusion.

A Gated Multimodal Fusion Network merges the embeddings of the two specialized models: the output of the ECG encoder and an echo-side embedding obtained from a video-based encoder for EchoNet-Dynamic/CAMUS instances or a tabular feature encoder for structured echocardiographic measurements where suitable paired records are available. The echo model's reduced-EF probability, predicted EF, and predictive uncertainty can also be used as additional fusion inputs, enabling the gating mechanism to weight echo evidence by its confidence. The learned gating mechanism decides the per-sample weighting for each modality, generating the final triage decision and an interpretable indication of modality reliance under domain shift. Fusion will be trained and tested using genuinely patient-paired ECG-echo records from MIMIC-IV-ECG [21] and MIMIC-IV-Echo [22], subject to verification of record linkage and availability, and compared with a rule-based fusion baseline and each single-modality model alone.

3.6 Model Compression.

The two final models are quantized and distilled and benchmarked for accuracy, latency and memory on standard versus constrained hardware.

3.7 Explainable Prototype and Evaluation.

Triage outputs, confidence scores, and Grad-CAM-style visual explanations are combined in a fully functional dashboard, validated in a formative usability pilot with PCPs. For echocardiography, triage is based on estimated EF: values above 50% are classified as Normal, values from 40% to 50% as Needs Review, and values below 40% as Urgent Referral. If the model's prediction is more uncertain than the permitted threshold, the case is sent for Needs Review no matter what the EF value is.

On this dashboard, an on-demand Referral Summary Generation function will be available after the relevant screening analysis is completed. An authenticated clinician will be able to review the results shown on the screen and trigger summary generation as a dedicated action in the UI. The summary will compile the relevant case metrics, the available findings from the ECG and echocardiography (as well as multimodal output where available), included confidence and uncertainty metrics, explainability sources, and the final triage category. The FastAPI backend will use a predefined template to create the summary and will store it with the respective case in Supabase. If only one modality has been analyzed, the summary will contain only the results of that modality, mentioning that the other modality is not available. This feature will not influence model Predict or change the original triage result, the clinician will, of course, have to review the summary before using it to communicate referrals.

3.8 Web Application Development and Deployment

The application shall use Supabase Authentication to support new-user registration, email verification, email/password sign-in, sign-out, and password recovery. New users shall register through the web application and complete the required email verification before accessing protected features. After successful sign-in, Supabase shall provide an authentication session and token that the frontend uses when communicating with the FastAPI backend. The backend shall validate the authentication token and independently enforce user and role-based authorization for protected operations.

Authentication: Users register and sign in using their email and password through Supabase Authentication. The system supports email verification, sign-out and password recovery. Users must complete email verification before accessing protected application features. Two application roles are defined: Clinician (upload and view their own cases and predictions) and Admin (manage users and application-level records). Authentication tokens are validated by the backend before processing protected requests.

Authorization: Access control is enforced at both the database and backend levels. Supabase Row-Level Security (RLS) policies restrict users to accessing only their authorized case records and associated files. The FastAPI service independently validates Supabase authentication tokens before accepting prediction requests. Administrative permissions are restricted to authorized users, and frontend screen restrictions alone are not considered sufficient security. Access to authenticated clinicians who have permission to the respective case to generate the referral summary and access the stored summary. The backend will check for case ownership or other authorization to retrieve case data, and may update the command store or hand back the created summary.

Data Management: Supabase PostgreSQL users profiles, pseudonymous case identifiers, permittable patient or case details, examination timestamps, input modality, processing status, model predictions, class probabilities, confidence scores, uncertainty estimates, explanation references, and generated referral summary records. The prototype's datasets of used case information will be de-identified or simulated, and will not involve collection of primary patient data. Each generated summary will be linked to its corresponding case and will remember its generation timestamp and associated analysis reference. Supabase Storage uploads ECG files, echocardiography videos, explanation output and summary documents if document export is implemented. File access is restricted by applying the appropriate access policies with Row-Level Security (RLS), and data deletion is handled by configurable retention rules. Supabase service role credentials and other application secrets are kept on the backend only and never exposed to the frontend. De-identified public datasets and simulated cases are used in the research and test phases.

Referral Summary Generation: The web application will have a quick-to-use Referral Summary Generation feature which will automatically activate as soon as at least one required cardiac screening analysis is completed successfully by an authenticated clinician. A case results dashboard will include a dedicated "Generate Referral Summary" action. When a clinician triggers the action, the React frontend will use an authenticated request containing the appropriate case reference to forward the request to the FastAPI backend. The worker in the backend will validate the user's authentication token and will warrant that the requested case ID is within the rights of the requesting clinician. The worker will then lookup the available pseudonymous patient or case information and outputs from completed analysis present in the Supabase PostgreSQL tables. These may contain ECG predictions, echocardiography findings and ejection fraction estimates, multimodal fusion results containing paired data, confidence scores, uncertainty estimates, explainability references, and the ultimate triage category. The newly acquired information will be organized into a structured referral summary by a deterministic function utilizing a predetermined template. The function will only use information for the selected case. The summary will

indicate an unavailable modality and will not create or make any presumptions of missing findings if a single modality was not examined. The summary will keep the original model predictions and triage outcome without diagnosing the patient themselves or modifying the screening results. The frontend will present the generated summary to the clinician for review in this view, and the backend will immediately store the summary (along with the generation timestamp and relevant analysis reference) as a record that is linked to the appropriate case in Supabase PostgreSQL. Clinicians will have the opportunity to review and potentially reference the returned summary in their referral communication. This will not be an automatically generated summary, run through a background scheduler, or will automatically consider an emergency case. PDF only available for the final application implementation.

Request Flow: (1) A new user signs up through the React web application, verifies their email, and logs in via Supabase Authentication to gain an authentication session and token. (2) An authenticated user begins a new case or accesses an existing case, provides the allowed pseudonymous case details, and uploads either the ECG file or the echocardiography video to Supabase Storage. (3) The frontend sends the analysis request to the FastAPI inference service with the authentication token and case reference tied to it. (4) The token is validated by the backend, after which the uploaded data is fetched and prepared, and the efficient ONNX models are executed for ECG classification, echocardiography analysis and multimodal fusion where paired data exist. (5) The predictions, confidence scores, uncertainty estimates, and explanation outputs are stored in the Supabase PostgreSQL, and an update is made to the transaction status and the case with the provided results and triage outcome as Normal, Needs Review or Urgent Referral. (6) Following the completion of the results review, the clinician may select "Generate Referral Summary." The frontend dispatches an authenticated request to FastAPI with the case reference chosen. (7) The backend validates that user has permission to read this case, obtains the case information available on the completed analysis outputs, and creates a structured summary using a prepopulated template. (8) The summary is saved in Supabase PostgreSQL against the respective case, and returned to the client for the clinician to read.

Deployment: The React frontend and FastAPI backend are deployed as separate application components, with the backend containerized to support consistent deployment and ONNX Runtime inference. Supabase Cloud provides the managed PostgreSQL database, authentication and file storage services. All application components communicate through secured HTTPS endpoints. Deployment instructions, environment configuration and required dependencies are documented to support reproducible deployment and future maintenance. The complete application is accessible through a standard web browser without requiring local model installation. An Authenticated endpoint for on demand referral summary generation from the FastAPI backend, while the React front end will provide the clinician-facing action and summary display. Generated summary records will be saved in Supabase PostgreSQL linked to their respective cases, and access will be managed via backend authorization and Row-Level Security (RLS) policies.

Application Testing: Application Testing. Tests cover user registration and authentication, email verification, role-based authorization, cross-user access rejection, file uploads, ECG and echocardiography prediction workflows, multimodal fusion, database and storage persistence, API error handling, deployment functionality, response time, and security. Referral Summary Generation New feature of the system will be tested to verify that only authorized clinicians can trigger the generation of summaries for cases they have access to; summaries contain all available case information, including outputs of completed analyses; and that model predictions and triage outcome of case and patient presented in the summary and the original modality data used for predictions are preserved. The test cases also confirm that the summary does not construct modality results that were not originally available. It will also make sure the proper summary was stored under the correct case and can be queried and accessed by the clinician for review. Usability assessment: using the System Usability Scale (SUS), task completion metrics, and summary-generation workflow performance evaluation (see Table 5).

System Architecture

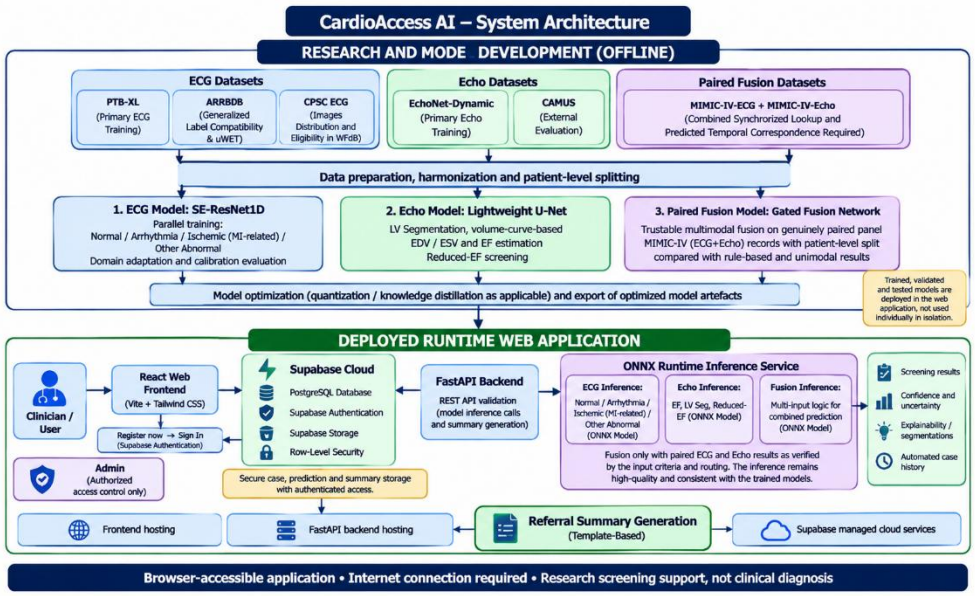


Figure 4: System Architecture

Diagram of the architecture of the deployed Web App and the independent Model-Development Pipeline, as illustrated above. A React front end uses HTTPS to talk to a FastAPI back end, where authentication is verified and model inference is orchestrated. The compressed ECG and echo models live in the model-serving layer, appropriate predictions are combined by the learned fusion layer with paired modality data when available. When finished screening results are examined, the authenticated clinical can request a structured referral summary via the frontend using an API call. The case metadata and analysis outputs currently in progress are lookup in Supabase PostgreSQL based on case id, the predefined template is used to render the summary which is then stored in Supabase PostgreSQL. Supabase also provides both authentication as well as storage of prediction results attached to each authorized user to their account. A reliable network connection is required for normal application behavior

- **User layer:** The clinician or nurse, the system's intended non-specialist end user.
- **Presentation layer:** A couldish-based front-end dashboard showing triage conclusions, confidence level, explainability overlays, case history, and a clinician-triggered "Generate Referral Summary" button with a review interface of summary.
- **Application layer:** The FastAPI backend layer for an authenticated request, that checks user permissions, orchestrates model inference, pulls case information, and earlier analysis results. Also generates structured referral summaries on clinician request thanks to a pre-built summary template. It then stores these summaries against their cases in Supabase for the clinician to review.
- **Model serving layer:** The compressed, ONNX-exported ECG and echo inference models, each running independently before their outputs are combined.
- **Fusion layer:** The learned fusion network producing the final triage decision from both modalities outputs.
- **Data layer:** Supabase Authentication & PostgreSQL hold authenticated user profiles, pseudonymous case information, prediction results, confidence and uncertainty estimates, explanation references, and referral summary records generated and indexed to cases. Supabase Storage keeps uploaded ECG and echocardiography files and generated explanation outputs, document storage is used for this if PDF export is enabled. All access is gated through user identity, superuser/backend authorization and Row-Level Security (both user and service role

tokens are in the backend and never exposed in the React front end).

- **Model-development pipeline:** Dataset preparation, baseline training, domain adaptation, active learning, and compression are completed during development. The final model artifacts are deployed behind the FastAPI inference service. The deployed application uses Supabase for authentication and persistent user-linked prediction storage.

The system is a deployable React web application with a FastAPI backend. It requires internet connectivity during normal use to reach the backend and Supabase services; training datasets and development compute are not required on the end-user device.

Software Development Methodology

Iterative Development - Scrum framework, with weekly sprints

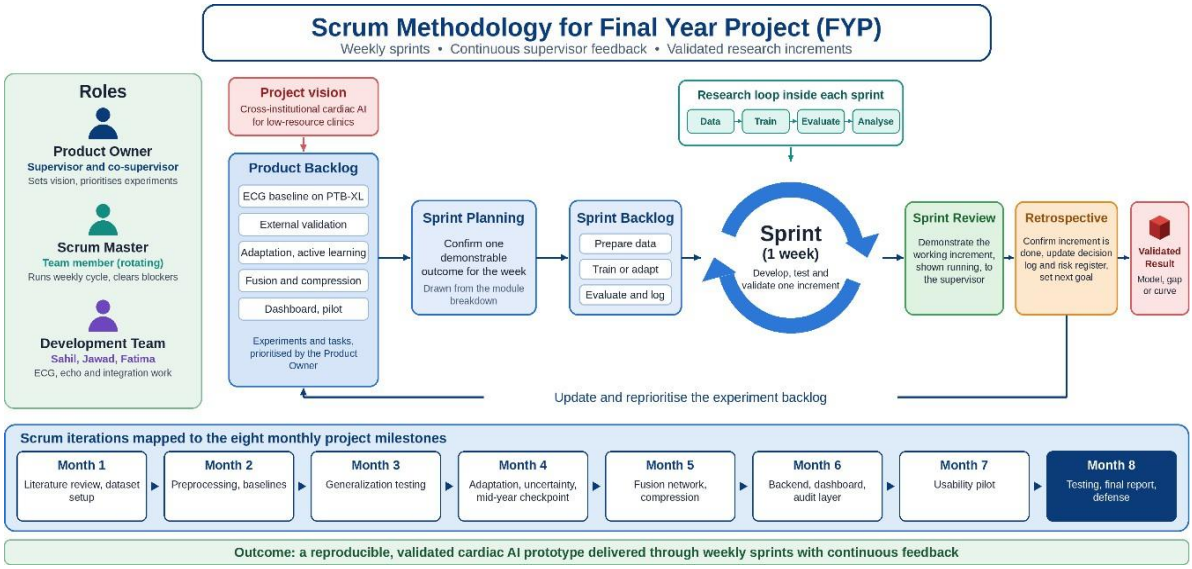


Figure 5: Software Development Methodology Diagram

Each week follows the same four-stage cycle:

5.1 Planning

(the team confirms the specific, demonstrable outcome for the week, drawn directly from the module breakdown for example, "ECG baseline model trained and evaluated on PTB-XL's held-out test set").

5.2 Development

(the assigned team member(s) build and test that specific increment).

5.3 Review

(a short session, ideally with the supervisor, where the working increment is demonstrated not described, actually shown running).

5.4 Retrospective

(The team confirms the increment is complete, updates the shared decision log and risk register, and sets the next week's goal accordingly). This weekly cadence is preferred over a monthly one because ML research work surfaces problems quickly (a failed training run, a preprocessing bug, a dataset format issue), and a month-long feedback loop would let such problems compound before being caught. A weekly cycle catches issues within days, matching the fast iteration speed that model training and evaluation actually require, while still rolling up cleanly into the eight monthly milestones already defined in the project timeline and workflow chart above.

Applications of the Project

6.1 Frontline triage in specialist-scarce clinics:

The system can also be used by a general physician or nurse in a rural or semirural Pakistani clinic to screen a patient's ECG and/or echocardiography data and receive a result (Normal, Needs Review, or Urgent Referral) without a cardiologist physically present. The diagnosis is made after analyzing the screening results. They can then produce a well-structured referral summary, which includes the available case info as well as analysis findings and triage outcomes of the screening. This summary can provide useful context to the specialist in cases when additional assessment is required. For echocardiography, triage is based on the perceived EF category as defined in Section 3.7.

6.2 Removing the specialist bottleneck:

It will prompt local healthcare workers that this may need millisecond hands to see if a specialist should see patients before risking patients to enormous and long trips to city hospitals for health care services. Its referral summary on an as-needed basis will align the available screening findings and triage information into a standardized format that clinicians can review before they use to back up their referral communication. Specialists would be able to better sort through the screening information they receive, and would be alerted to cases that needed more awaiting clinical evaluation.

6.3 Web-based access

The application is accessed through a web browser and requires a reliable internet connection to communicate with the deployed FastAPI backend and Supabase services. This makes internet connectivity a deployment requirement for normal operation.

6.4 Undergraduate / graduate medical & nursing training:

The explainability outputs of the system (Grad-CAM style visual attention overlays) enable learners to see their own interpretation of an ECG / echo case compared with the model's regions of interest, serving as a diagnostic training tool in environments where expert guidance is scarce to none.

6.5 A reusable benchmark for future low-resource cardiac AI research:

The project presents a general methodology and reference results that can be leveraged for future student or research projects in this area (such as CPEIC/Multan dataset lineage) by quantifying generalization loss, adaptation recovery, calibration behaviors, hardware feasibility in two modalities, thereby explicitly capturing these factors.

Project Management

7.1 Workflow Chart

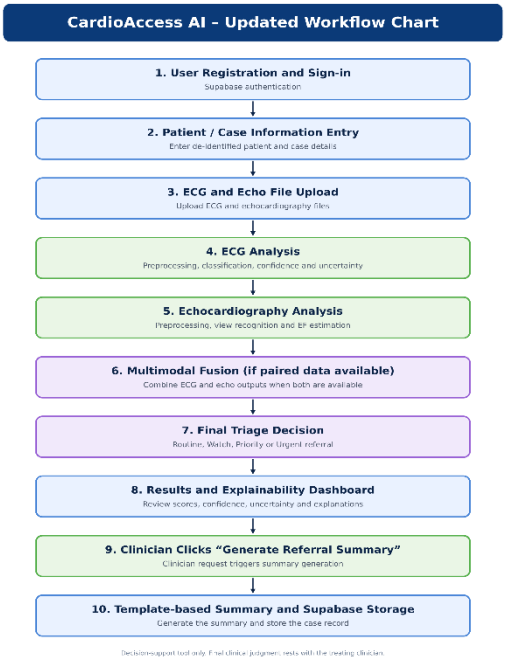


Figure 6: Workflow Chart

7.2 Project Breakdown

Table 2: Project Breakdown Table

Phase	Sahil Kumar	Jawad Ali	Fatima
Literature Review	✓	✓	✓
Dataset Acquisition & Setup	✓	✓	✗
ECG Preprocessing	✓	✗	✗
Echo Preprocessing	✓	✗	✗
ECG Baseline Model Training	✓	✗	✓
Echo Baseline Model Training	✓	✗	✓
ECG Generalization Testing	✓	✓	✗
Echo Generalization Testing	✓	✓	✗
Domain Adaptation (ECG)	✓	✗	✓
Domain Adaptation (Echo)	✓	✗	✓
Uncertainty Quantification	✓	✗	✓
Active Learning Integration	✓	✗	✓
Multimodal Fusion Network	✓	✓	✓
Model Compression	✗	✓	✗
Backend	✓	✓	✗
Frontend Dashboard	✗	✓	✓
Usability Pilot	✓	✓	✗
Testing & Refinement	✓	✓	✓
Documentation & Final Report	✓	✓	✓

Supabase Cloud Setup & Configuration	✓	✓	✕
PostgreSQL Database Design & Integration	✓	✓	✕
User Authentication & Role-Based Access Control	✓	✓	✕
ECG & Echo File Upload and Supabase Storage Integration	✓	✓	✕
Frontend–Backend API Integration	✓	✓	✓
ECG, Echo & Fusion Model Integration with Backend	✓	✕	✓
Cloud Deployment & Production Configuration	✓	✓	✕
Referral Summary Data Mapping and Template Design	✓	✓	✓
Backend Referral Summary Generation	✕	✕	✓
Frontend Summary Generation and Review Interface	✕	✕	✓
Summary Storage and Case Linking in Supabase	✓	✕	✓
Referral Summary Functional and Security Testing	✓	✓	✓

7.3 Gantt Chart

The project schedule is also visualized through a Gantt chart for clarity:

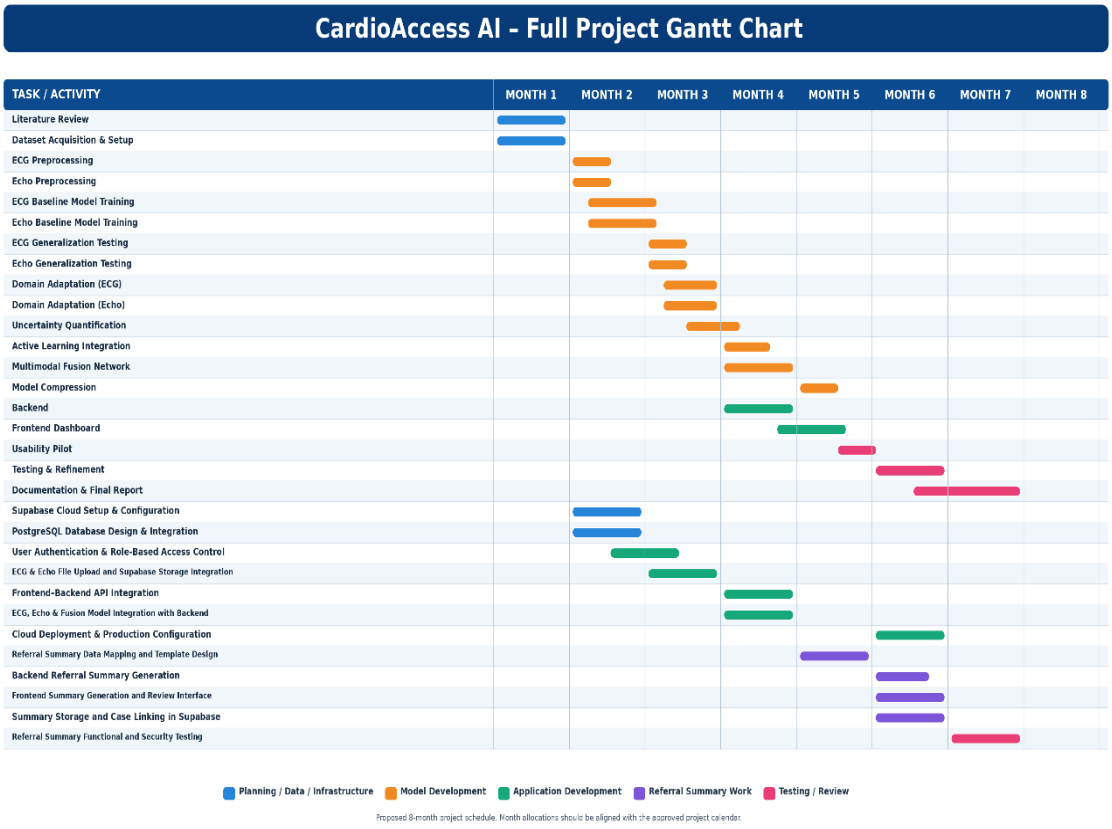


Figure 7: Gantt Chart

7.4 Roles & Responsibilities

7.4.1 Sahil Kumar: Data & ECG Modeling

- ECG dataset acquisition, preprocessing, and patient-level splitting.
- ECG baseline model training and generalization testing.
- ECG domain adaptation and active learning integration.
- Support pseudonymous case data mapping, database consistency, and validation of information included in generated referral summaries.

7.4.2 Jawad Ali: Echo Modeling & Compression

- Echo dataset acquisition and format conversion (CAMUS, EchoNet-Dynamic).
- Echo baseline model training and generalization testing.
- Model compression and hardware benchmarking (both modalities).

7.4.3 Fatima: Integration & Product

- Backend development (FastAPI, authentication, and Supabase integration).
- Frontend dashboard development (React.js).
- Usability pilot coordination and integration of authentication and user-linked prediction storage.
- Create the on-demand Referral Summary Generation feature in the FastAPI backend (case info retrieval, completed analysis outputs, template or code based generation, storing in Supabase).
- Implement the clinician-facing summary generation action and summary review interface in the React UI (bootstrap and authentication).

7.4.4 Shared Responsibilities:

- Literature review and weekly decision log updates.
- Multimodal fusion network (requires both modality tracks joint work).
- Proposal, mid-year report, final report, and defense preparation.
- Validate the completeness, consistency, access control, and usability of generated referral summaries using de-identified or simulated cases.

7.5 Tools & Technologies Required

Table 3: Tools & Technologies

Category	Tool/Technology	Purpose
Core programming language (ML)	Python 3.x	Model development, data pipelines, backend logic
Core programming language (frontend)	JavaScript	React application development
ML/DL framework	PyTorch	Training the ECG and echo models, domain adaptation, fusion network
Model export/inference	ONNX Runtime	Deploying compressed models for low-power inference

Explainability	Captum (PyTorch) / custom Grad-CAM implementation	Generating explainability heatmaps for ECG and echo predictions
Uncertainty quantification	PyTorch (Monte Carlo Dropout), custom ensemble training scripts	Generating model confidence estimates
Signal processing (ECG)	wfdb (Python package), NumPy, SciPy	Parsing PTB-XL/ARRDB signal data, filtering, normalization
Image/video processing (Echo)	OpenCV, NiBabel	Frame extraction from EchoNet-Dynamic video, NIfTI parsing for CAMUS
Backend framework	FastAPI (Python)	REST API serving model predictions, managing case information, validating authenticated requests, and generating on-demand template-based referral summaries.
Database	Supabase Auth + PostgreSQL	User authentication, secure storage of pseudonymous case information and prediction results, and persistent storage of generated referral summary records linked to their corresponding cases using Row-Level Security.
Frontend framework	React	Clinician-facing dashboard displaying triage results, explainability overlays, user-linked case history, and the on-demand referral summary generation and review interface.
Frontend build tooling	Vite	Fast development server and build pipeline for the React app
Frontend styling	Tailwind CSS	Utility-first styling for the dashboard UI
Frontend charts/visualization	Recharts or Chart.js	Displaying confidence scores, calibration curves, and case history in the dashboard
API communication	Axios / Fetch API	Connecting the React frontend to the FastAPI backend for authenticated prediction requests, case operations, and on-demand referral summary generation and retrieval.
Compute environment	Google Colab (Pro) & Kaggle Notebook	GPU-accelerated model training, particularly for the echo video model
Version control	Git, GitHub	Feature-branch workflow across the three-person team

Model compression	PyTorch quantization toolkit, ONNX Runtime quantization tools	Post-training quantization and knowledge distillation
Testing	pytest	Backend and data pipeline unit/integration testing
Documentation	Microsoft Word	Formal proposal, mid-year report, and thesis document
Collaboration	Google Docs / Notion	Shared weekly decision log and risk register
Frontend hosting	Vercel	Deploying the React web application
Backend hosting	Render/Railway	Hosting the FastAPI inference service
Cloud database and authentication	Supabase Cloud	PostgreSQL, authentication and persistent prediction storage

7.6 Project Feasibility

1. **Technical feasibility:** The project uses seven datasets across the ECG and echo tracks: PTB-XL, ARRDB, EchoNet-Dynamic, CAMUS, MIMIC-IV-ECG, MIMIC-IV-Echo, and a locally sourced ECG dataset. MIMIC-IV access is credentialed and subject to the applicable data-use requirements. The application is built with standard software/AI tools and does not require proprietary physical hardware. But how this ends up working technologically is as expected; things can be fulfilled because they have already relied on the FastAPI backend, React frontend, and Supabase PostgreSQL infrastructure that are already set up. Additionally, there is no necessity for an extra diagnostic model, scheduler service, or hardware, it utilizes a preset template and existing case and analysis data.

2. **Economic feasibility:** Since the FYP is not hardware-dependent, students are not required to buy any physical hardware, unlike in hardware-dependent FYPs, thus the cost is negligible and is just for (optional) cloud compute (see Section 3).

3. **Operational feasibility:** In Section 2.4, we propose that the three-person team divide responsibility across two modality tracks plus an integration role, aligning with the natural module boundaries of the project. The referral summary functionality will be implemented as part of the existing application integration work and will use the team's established backend, frontend, and database responsibilities.

4. **Ethical feasibility:** The project relies solely on public, de-identified datasets for its core research; it collects no primary human-subjects data, and the usability pilot will be conducted using informed, voluntary participation with no clinical claims made to participants. Referral summaries will be created and analysed using de-identified or synthetic case information. Summaries are to be provided to clinicians as referral-support documents for clinicians to consider, not as independent clinical diagnoses or treatment recommendations.

Costing

Table 4: Costing

Category	PKR
Compute Colab (Pro)	0
Public Datasets	0
Supabase Cloud (PostgreSQL, Authentication & Storage)	To be Estimated
Cloud Storage / Backup	150,000
Contingency (ECG Data)	1,00,000
Domain hosting Demo	40,000
Misc. Supplies	15,000
Claude Code Premium	1,00,000

Total Estimated Cost: Rs. 4,05,000

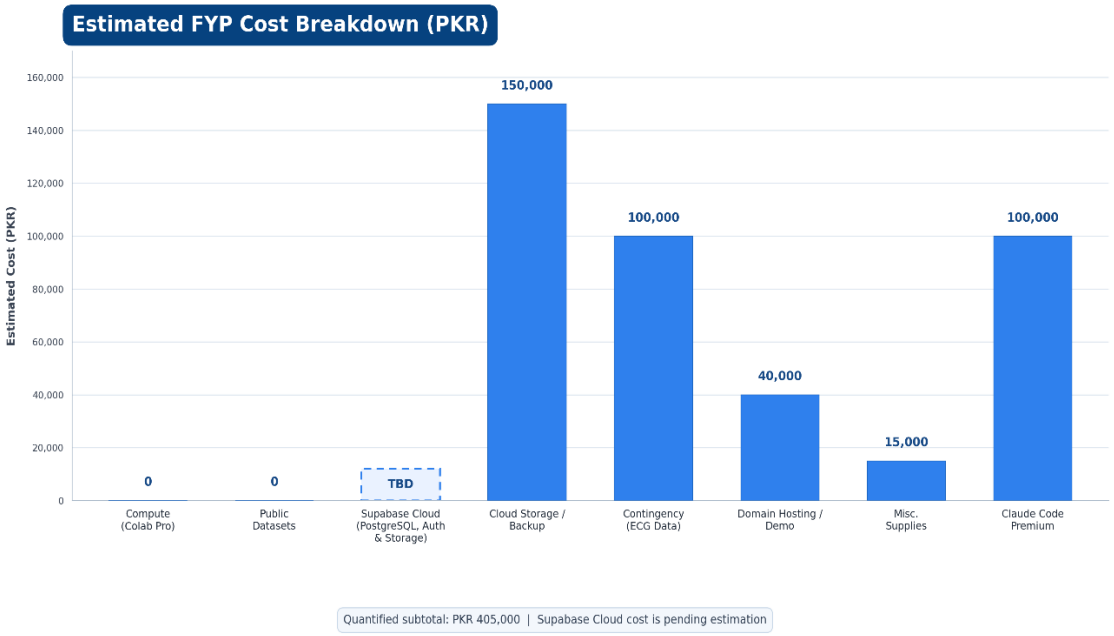


Figure 8: Cost Estimation

Evaluation Metrics

Table 5: Evaluation Metrics

#	Component	Metric(s)	What It Demonstrates
1	ECG baseline model	AUC, sensitivity, specificity (per class)	In-distribution diagnostic performance on PTB-XL
2	Echo segmentation	Dice coefficient, IoU (end-diastole and end-systole frames)	Accuracy of LV segmentation on EchoNet-Dynamic
3	Echo EF estimation	EF MAE, RMSE, R^2, Bland–Altman agreement	Accuracy of the ejection fraction value against the labeled EF
4	Echo reduced-EF screening	AUC, sensitivity, specificity, PR-AUC (EF < 50% and ≤ 40%)	Ability to flag reduced function; PR-AUC is used because the classes are imbalanced (about 22% reduced at EF < 50)
5	Cross-institutional generalization	Absolute and percentage drop in the above metrics (source dataset vs. external dataset)	Quantifies the real-world generalization gap for both modalities
6	Domain adaptation (fine-tuning)	Recovery curve: performance recovered at 5%, 10%, and 25% target-data budgets	How much labeled local data is needed to close the generalization gap
7	Domain adaptation (CORAL)	Recovery achieved with zero target labels	Feasibility of adaptation when no local labels are available at all
8	Active learning vs. random sampling	Recovery curve comparison at matched data budgets	Whether uncertainty-guided sample selection improves labeling efficiency
9	Confidence calibration	Expected Calibration Error (ECE), Brier score, reliability diagrams, in-distribution vs. out-of-distribution; ECG: class probabilities, echo: reduced-EF probability	Whether model confidence remains trustworthy after domain shift
10	Multimodal fusion	Accuracy/AUC of learned fusion vs. rule-based fusion vs. each single-modality model	Whether joint ECG–echo reasoning improves robustness beyond either modality alone
11	Model compression	Accuracy retained (%), inference latency, memory footprint (standard vs. constrained hardware)	Real-world deployability on low-power, primary-care-representative hardware

12	Subgroup fairness	Performance stratified by age and sex, before and after adaptation	Whether generalization or adaptation introduces demographic performance disparities
13	Usability pilot	Task completion time, System Usability Scale (SUS) score, qualitative trust feedback	Practical usability and perceived trustworthiness among primary care provider volunteers
14	Clinical screening utility	Negative predictive value (NPV), positive predictive value (PPV), false-negative rate (FNR), and decision-curve analysis (if feasible)	Evaluates whether the system can support practical screening decisions while considering missed cases and unnecessary referrals.
15	Referral Summary Generation	Information completeness, consistency with stored analysis results, correct triage representation, successful generation rate, and task-completion time	Whether the application generates complete and consistent referral summaries from available case information and analysis outputs, preserves the original screening results, and supports an efficient clinician review workflow.

Future Work

Beyond the seven gaps addressed directly, this work also surfaces three related questions for future research: whether a lightweight, adaptation-based approach of the kind developed here offers a meaningfully different robustness-for-compute trade-off compared to large-scale foundation models such as those evaluated in CardioBench [16]; whether lightweight adaptation is a more practical solution than federated learning specifically for clinics that lack the connectivity or infrastructure required for multi-institutional data-sharing coordination [14][15]; and whether synthetic or generative augmentation of scarce target-domain data could further reduce the amount of real local data required for effective adaptation. Each of these questions builds directly on the empirical findings this project is expected to produce.

Conclusion

CardioAccess AI fills a well-defined and evidence-supported gap in cardiac artificial intelligence: the sustained disparity between the performance of AI models in controlled research environments and how trustworthy they would be in an under-resourced clinic. In this work, by quantifying cross-institutional generalization between two cardiac modalities, utilizing lightweight and label-efficient adaptation methods, and demonstrating the computational feasibility and usability of the resulting system, we provide both a rigorous empirical contribution as well as a functional, explainable prototype. The work is realistically scoped to be completed by a three-person team in two semesters, is based on public and locally available datasets, and has immediate applicability to the delivery of healthcare in Pakistan and similar resource-poor environments worldwide. We will also be putting a prototype of an on-demand Referral Summary Generation feature to the demo; this allows certified clinicians to view and organize all Cardiac screening findings in structured summaries saved along with their own cases.

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